

EXT 2: Complex Numbers (Ext2), N2 Using Complex Numbers (Ext2) Solving Equations with Complex Numbers

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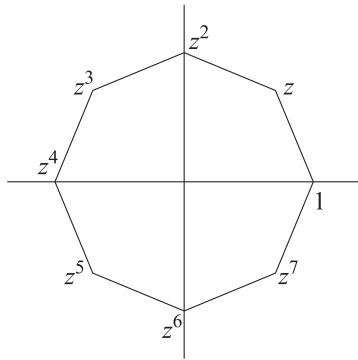
Exam Equivalent Time: 69 minutes (based on HSC allocation of 1.5 minutes approx. per mark)



Questions

1. Complex Numbers, EXT2 N2 2017 HSC 1 MC

The complex number z is chosen so that $1, z, \dots, z^7$ form the vertices of the regular polygon shown.



Which polynomial equation has all of these complex numbers as roots?

- A. $x^7 - 1 = 0$
- B. $x^7 + 1 = 0$
- C. $x^8 - 1 = 0$
- D. $x^8 + 1 = 0$

2. Complex Numbers, EXT2 N2 2018 HSC 6 MC

Which complex number is a 6th root of i ?

- A. $-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$
- B. $-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$
- C. $-\sqrt{2} + \sqrt{2}i$
- D. $-\sqrt{2} - \sqrt{2}i$

3. Complex Numbers, EXT2 N2 2016 HSC 12c

Let $z = \cos \theta + i \sin \theta$.

i. By considering the real part of z^4 , show that $\cos 4\theta$ is

$$\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta. \quad (2 \text{ marks})$$

ii. Hence, or otherwise, find an expression for $\cos 4\theta$ involving only powers of $\cos \theta$. (1 mark)

4. Complex Numbers, EXT2 N2 2011 HSC 2c

Find, in modulus-argument form, all solutions of $z^3 = 8$. (2 marks)

5. Complex Numbers, EXT2 N2 2011 HSC 2d

i. Use the binomial theorem to expand $(\cos \theta + i \sin \theta)^3$. (1 mark)

ii. Use de Moivre's theorem and your result from part (i) to prove that

$$\cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta. \quad (3 \text{ marks})$$

iii. Hence, or otherwise, find the smallest positive solution of

$$4 \cos^3 \theta - 3 \cos \theta = 1. \quad (2 \text{ marks})$$

6. Complex Numbers, EXT2 N2 EQ-Bank 1

$z = \sqrt{2}e^{\frac{i\pi}{15}}$ is a root of the equation $z^5 = \alpha(1 + i\sqrt{3})$, $\alpha \in \mathbb{R}$.

i. Express $1 + i\sqrt{3}$ in exponential form. (2 marks)

ii. Find the value of α . (1 mark)

iii. Find the other 4 roots of the equation in exponential form. (3 marks)

7. Complex Numbers, EXT2 N2 2018 HSC 15b

i. Use De Moivre's theorem and the expansion of $(\cos \theta + i \sin \theta)^8$ to show that

$$\begin{aligned}\sin 8\theta &= \binom{8}{1} \cos^7 \theta \sin \theta - \binom{8}{3} \cos^5 \theta \sin^3 \theta \\ &\quad + \binom{8}{5} \cos^3 \theta \sin^5 \theta - \binom{8}{7} \cos \theta \sin^7 \theta \quad (2 \text{ marks})\end{aligned}$$

ii. Hence, show that

$$\frac{\sin 8\theta}{\sin 2\theta} = 4(1 - 10 \sin^2 \theta + 24 \sin^4 \theta - 16 \sin^6 \theta). \quad (3 \text{ marks})$$

8. Complex Numbers, EXT2 N2 2006 HSC 2c

Find, in modulus-argument form, all solutions of $z^3 = -1$. (2 marks)

9. Complex Numbers, EXT2 N2 2009 HSC 2e

i. Find all the 5th roots of -1 in modulus-argument form. (2 marks)

ii. Sketch the 5th roots of -1 on an Argand diagram. (1 mark)

10. Complex Numbers, EXT2 N2 2020 SPEC1 3

Find the cube roots of $\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$. Express your answers in modulus-argument form. (3 marks)

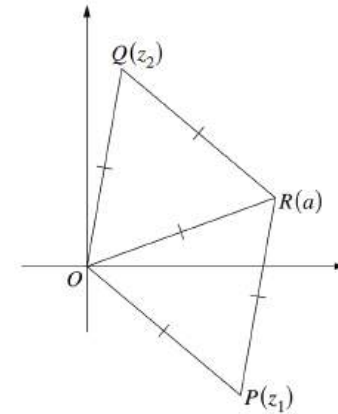
11. Complex Numbers, EXT2 N2 SM-Bank 1

Find all solutions for z , in exponential form, given $z^4 = -2\sqrt{3} - 2i$. (3 marks)

12. Complex Numbers, EXT2 N2 SM-Bank 8

If $(x + iy)^3 = e^{-\frac{i\pi}{2}}$, $x, y \in \mathbb{R}$, find a solution in the form $x + iy$, $x, y \neq 0$. (2 marks)

13. Complex Numbers, EXT2 N2 2007 HSC 2d



The points P, Q and R on the Argand diagram represent the complex numbers z_1, z_2 and a respectively.

The triangles OPR and OQR are equilateral with unit sides, so $|z_1| = |z_2| = |a| = 1$.

Let $\omega = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$.

i. Explain why $z_2 = \omega a$. (1 mark)

ii. Show that $z_1 z_2 = a^2$. (1 mark)

iii. Show that z_1 and z_2 are the roots of $z^2 - az + a^2 = 0$. (2 marks)

14. Complex Numbers, EXT2 N2 2014 HSC 15b

i. Using de Moivre's theorem, or otherwise, show that for every positive integer n ,

$$(1 + i)^n + (1 - i)^n = 2(\sqrt{2})^n \cos \frac{n\pi}{4}. \quad (2 \text{ marks})$$

ii. Hence, or otherwise, show that for every positive integer n divisible by 4,

$$\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots + \binom{n}{n} = (-1)^{\frac{n}{4}} (\sqrt{2})^n \quad (3 \text{ marks})$$

Worked Solutions

1. Complex Numbers, EXT2 N2 2017 HSC 1 MC

$P(x)$ has 8 separate roots.

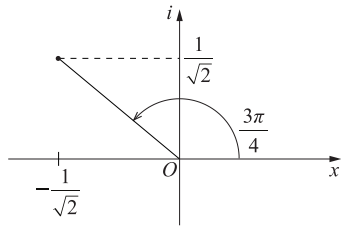
$\therefore$ Must be of degree at least 8.

Since 1 is also a root,

$$\Rightarrow C$$

2. Complex Numbers, EXT2 N2 2018 HSC 6 MC

Consider option A:



$$\left| -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right| = \sqrt{\left(-\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = 1$$

$$\arg(z) = \frac{3\pi}{4}$$

$$z = 1 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$z^6 = \cos \frac{18\pi}{4} + i \sin \frac{18\pi}{4} \quad (\text{De Moivre})$$

$$= i$$

$$\Rightarrow A$$

3. Complex Numbers, EXT2 N2 2016 HSC 12c

i. $z = \cos \theta + i \sin \theta$

$$\begin{aligned} z^4 &= (\cos \theta + i \sin \theta)^4 \\ &= \cos^4 \theta + 4 \cos^3 \theta \cdot (i \sin \theta) + 6 \cos^2 \theta \cdot (i \sin \theta)^2 + \\ &\quad 4 \cos \theta \cdot (i \sin \theta)^3 + (i \sin \theta)^4 \\ &= \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - \\ &\quad 4i \cos \theta \sin^3 \theta + \sin^4 \theta \end{aligned}$$

$$z^4 = \cos 4\theta + i \sin 4\theta \quad (\text{by De Moivre})$$

Equating real parts:

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

... as required

ii. $\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$

$$\begin{aligned} &= \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta \\ &= 8 \cos^4 \theta - 8 \cos^2 \theta + 1 \end{aligned}$$

4. Complex Numbers, EXT2 N2 2011 HSC 2c

$$z^3 = 8$$

$$z^3 = 8[\cos(2k\pi) + i \sin(2k\pi)], \quad k = 0, \pm 1, \pm 2$$

$$z = 2 \left[\cos\left(\frac{2k\pi}{3}\right) + i \sin\left(\frac{2k\pi}{3}\right) \right] \quad (\text{De Moivre})$$

When $k = 0$,

$$z = 2(\cos 0 + i \sin 0)$$

$$= 2\text{cis } 0$$

When $k = 1$,

$$z = 2 \left[\cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) \right]$$

$$= 2\text{cis } \frac{2\pi}{3}$$

When $k = -1$,

$$z = 2 \left[\cos\left(\frac{-2\pi}{3}\right) + i \sin\left(\frac{-2\pi}{3}\right) \right]$$

$$= 2\text{cis}\left(\frac{-2\pi}{3}\right)$$

5. Complex Numbers, EXT2 N2 2011 HSC 2d

i. $(\cos \theta + i \sin \theta)^3$

$$= \sum_{k=0}^3 {}^3C_k (\cos \theta)^{3-k} (i \sin \theta)^k$$

$$= \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3$$

$$= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$$

ii. Using De Moivre's Theorem

$$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$$

Equate real parts

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$$

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$$

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$4 \cos^3 \theta = \cos 3\theta + 3 \cos \theta$$

$$\therefore \cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta \quad \dots \text{as required}$$

iii. If $4 \cos^3 \theta - 3 \cos \theta = 1$

$$\Rightarrow \cos 3\theta = 1 \quad (\text{from part (ii)})$$

$$3\theta = 2k\pi$$

$$\therefore \theta = \frac{2k\pi}{3}$$

$\therefore$ Smallest positive solution occurs when

$$\theta = \frac{2\pi}{3} \quad (\text{i.e. when } k = 1)$$

6. Complex Numbers, EXT2 N2 EQ-Bank 1

i. $\beta = 1 + i\sqrt{3}$

$$|\beta| = \sqrt{1 + (\sqrt{3})^2} = 2$$

$$\arg(\beta) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$$

$$\beta = 2e^{\frac{i\pi}{3}}$$

ii. $z = \sqrt{2}e^{\frac{i\pi}{15}}$

$$\begin{aligned} z^5 &= \left(\sqrt{2}e^{\frac{i\pi}{15}}\right)^5 \\ &= (\sqrt{2})^5 e^{\frac{i\pi}{15} \times 5} \\ &= 4\sqrt{2}e^{\frac{i\pi}{3}} \end{aligned}$$

iii. $\arg(z^5) = \frac{\pi}{3} + 2k\pi, \quad k = 0, \pm 1, \pm 2, \dots$

$$\arg(z) = \frac{\pi}{15} + \frac{2k\pi}{5}$$

$$k = 1: \arg(z) = \frac{\pi}{15} + \frac{2\pi}{5} = \frac{11\pi}{15}$$

$$k = -1: \arg(z) = \frac{\pi}{15} - \frac{2\pi}{5} = -\frac{\pi}{3}$$

$$k = 2: \arg(z) = \frac{\pi}{15} + \frac{4\pi}{5} = \frac{13\pi}{15}$$

$$k = -2: \arg(z) = \frac{\pi}{15} - \frac{4\pi}{5} = -\frac{11\pi}{15}$$

$\therefore$ 4 other roots are:

$$e^{\frac{i11\pi}{15}}, e^{-\frac{i\pi}{3}}, e^{\frac{i13\pi}{15}}, e^{-\frac{i11\pi}{15}}$$

7. Complex Numbers, EXT2 N2 2018 HSC 15b

i. By De Moivre

$$\cos \theta + i \sin \theta^8 = \cos 8\theta + i \sin 8\theta \dots (*)$$

Using Binomial Expansion

$$\begin{aligned} (\cos \theta + i \sin \theta)^8 &= \cos^8 \theta + \binom{8}{1} \cos^7 \theta \cdot i \sin \theta + \binom{8}{2} \cos^6 \theta \cdot i^2 \sin^2 \theta \\ &\quad + \binom{8}{3} \cos^5 \theta \cdot i^3 \sin^3 \theta + \binom{8}{4} \cos^4 \theta \cdot i^4 \sin^4 \theta + \binom{8}{5} \cos^3 \theta \cdot i^5 \sin^5 \theta \\ &\quad + \binom{8}{6} \cos^2 \theta \cdot i^6 \sin^6 \theta + \binom{8}{7} \cos \theta \cdot i^7 \sin^7 \theta + i^8 \sin^8 \theta \end{aligned}$$

Equating imaginary parts of the expansion equation (*):

$$\begin{aligned} i \sin 8\theta &= \binom{8}{1} \cos^7 \theta \cdot i \sin \theta + \binom{8}{3} i \cos^5 \theta \cdot i^3 \sin^3 \theta \\ &\quad + \binom{8}{5} \cos^3 \theta \cdot i^5 \sin^5 \theta + \binom{8}{7} \cos \theta \cdot i^7 \sin^7 \theta \\ \therefore \sin 8\theta &= \binom{8}{1} \cos^7 \theta \sin \theta - \binom{8}{3} \cos^5 \theta \sin^3 \theta \\ &\quad + \binom{8}{5} \cos^3 \theta \sin^5 \theta - \binom{8}{7} \cos \theta \sin^7 \theta \end{aligned}$$

ii. $\begin{aligned} \sin 8\theta &= 8 \cos^7 \theta \sin \theta - 56 \cos^5 \theta \sin^3 \theta + 56 \cos^3 \theta \sin^5 \theta - 8 \cos \theta \sin^7 \theta \\ &= 2 \sin \theta \cos \theta (4 \cos^6 \theta - 28 \cos^4 \theta \sin^2 \theta + 28 \cos^2 \theta \sin^4 \theta - 4 \sin^6 \theta) \end{aligned}$

$$\begin{aligned} \therefore \frac{\sin 8\theta}{\sin 2\theta} &= 4 \cos^6 \theta - 28 \cos^4 \theta \sin^2 \theta + 28 \cos^2 \theta \sin^4 \theta - 4 \sin^6 \theta \\ &= 4(1 - \sin^2 \theta)^3 - 28(1 - \sin^2 \theta)^2 \sin^2 \theta + 28(1 - \sin^2 \theta) \sin^4 \theta - 4 \sin^6 \theta \\ &= 4(1 - 3 \sin^2 \theta + 3 \sin^4 \theta + \sin^6 \theta) - 28 \sin^2 \theta (1 - 2 \sin^2 \theta + \sin^4 \theta) \\ &\quad + 28 \sin^4 \theta (1 - \sin^2 \theta) - 4 \sin^6 \theta \\ &= 4 - 40 \sin^2 \theta + 96 \sin^4 \theta - 56 \sin^6 \theta \\ &= 4(1 - 10 \sin^2 \theta + 24 \sin^4 \theta - 16 \sin^6 \theta) \end{aligned}$$

8. Complex Numbers, EXT2 N2 2006 HSC 2c

$$z^3 = -1$$

$$-1 = \text{cis}(\pi + 2k\pi)$$

$$\therefore z = \text{cis} \frac{\pi + 2k\pi}{3} \quad (\text{De Moivre})$$

When $k = 0$

$$z = \text{cis} \frac{\pi}{3}$$

When $k = 1$

$$z = \text{cis} \pi = -1$$

When $k = -1$

$$z = \text{cis} -\frac{\pi}{3}$$

$\therefore$ The 3 solutions to $z^3 = -1$ are

$$z = \text{cis} \frac{\pi}{3}, \quad -1, \quad \text{cis} -\frac{\pi}{3}$$

MARKER'S COMMENT: Another successful solution strategy was graphical, using the unit circle and three equal angles generated from π .

9. Complex Numbers, EXT2 N2 2009 HSC 2e

i. $z = \cos \theta + i \sin \theta$

$$z^5 = \cos 5\theta + i \sin 5\theta = -1, \quad (\text{De Moivre})$$

$$\therefore \cos 5\theta = -1$$

$$5\theta = \pi, 3\pi, 5\pi, 7\pi, 9\pi$$

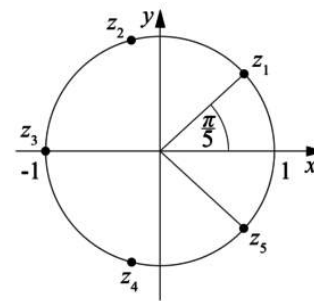
$$\theta = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5}$$

$\therefore$ The roots are

$$z_1 = \text{cis} \frac{\pi}{5}, \quad z_2 = \text{cis} \frac{3\pi}{5}, \quad z_3 = \text{cis} \pi = -1,$$

$$z_4 = \text{cis} \frac{7\pi}{5}, \quad z_5 = \text{cis} \frac{9\pi}{5}$$

ii.



10. Complex Numbers, EXT2 N2 2020 SPEC1 3

$$z^3 = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i = \text{cis}\left(-\frac{\pi}{4}\right) = \text{cis}\left(-\frac{\pi}{4} + 2\pi k\right), k \in \mathbb{Z}$$

By De Moivre:

$$z = \text{cis}\left(-\frac{\pi}{12} + \frac{2\pi k}{3}\right)$$

$$\text{If } k = 1, z_1 = \text{cis}\left(-\frac{\pi}{12} + \frac{2\pi}{3}\right) = \text{cis}\left(\frac{7\pi}{12}\right)$$

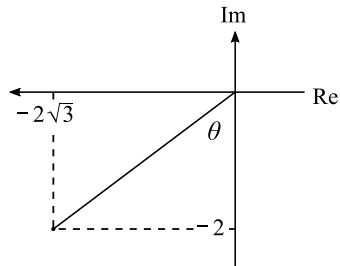
$$\text{If } k = 0, z_2 = \text{cis}\left(-\frac{\pi}{12}\right)$$

$$\text{If } k = -1, z_3 = \text{cis}\left(-\frac{\pi}{12} - \frac{2\pi}{3}\right) = \text{cis}\left(\frac{-3\pi}{4}\right)$$

11. Complex Numbers, EXT2 N2 SM-Bank 1

Convert z^4 to Mod/Arg:

$$|z^4| = \sqrt{(2\sqrt{3})^2 + 2^2} = 4$$



$$\tan \theta = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

$$\theta = \frac{\pi}{3}$$

$$\arg(z^4) = -\left(\frac{\pi}{2} + \frac{\pi}{3}\right) = -\frac{5\pi}{6}$$

By De Moivre:

$$|z| = \sqrt[4]{4} = \sqrt{2}$$

$$\arg(z) = -\frac{5\pi}{24} + \frac{k\pi}{2}, \quad k = 0, \pm 1, \pm 2$$

$$k = 0: \arg(z) = -\frac{5\pi}{24}$$

$$k = 1: \arg(z) = -\frac{5\pi}{24} + \frac{\pi}{2} = \frac{7\pi}{24}$$

$$k = -1: \arg(z) = -\frac{5\pi}{24} - \frac{\pi}{2} = \frac{-17\pi}{24}$$

$$k = 2: \arg(z) = -\frac{5\pi}{24} + \pi = \frac{19\pi}{24}$$

$$\therefore \text{Solutions are: } \sqrt{2}e^{-\frac{17\pi}{24}i}, \sqrt{2}e^{\frac{-5\pi}{24}i}, \sqrt{2}e^{\frac{7\pi}{24}i}, \sqrt{2}e^{\frac{19\pi}{24}i}$$

12. Complex Numbers, EXT2 N2 SM-Bank 8

$$\text{Let } (x + iy) = \cos \theta + i \sin \theta$$

$$(x + iy)^3 = \cos(3\theta) + i \sin(3\theta)$$

$$e^{-\frac{i\pi}{2}} = \cos\left(\frac{-\pi}{2}\right) + i \sin\left(\frac{-\pi}{2}\right)$$

Equating real parts:

$$3\theta = \frac{-\pi}{2}$$

$$\theta = \frac{-\pi}{6}$$

$$\begin{aligned} \therefore x + iy &= \cos\left(\frac{-\pi}{6}\right) + i \sin\left(\frac{-\pi}{6}\right) \\ &= \frac{\sqrt{3}}{2} - i\frac{1}{2} \end{aligned}$$

COMMENT: Read the question fully and save time - only one solution is required here.

13. Complex Numbers, EXT2 N2 2007 HSC 2d

i. Since $\triangle ORQ$ is equilateral, each angle is $\frac{\pi}{3}$ radians.

From $R(a)$:

$Q(z_2)$ is an anticlockwise rotation through $\frac{\pi}{3}$.

$$\therefore z_2 = a \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = \omega a.$$

ii. Solution 1

Similarly, $a = z_1 \omega$

$$z_1 = \frac{a}{\omega}$$

$$\begin{aligned} \therefore z_1 z_2 &= \frac{a}{\omega} \times \omega a \\ &= a^2 \end{aligned}$$

Solution 2

$P(z_1)$ is a clockwise rotation of $R(a)$ through $\frac{\pi}{3}$.

$$\therefore z_1 = \bar{\omega} a.$$

$$\therefore z_1 z_2 = \bar{\omega} a \times \omega a$$

$$\begin{aligned} &= a^2 \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right) \times \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \\ &= a^2 \left(\cos^2 \frac{\pi}{3} + \sin^2 \frac{\pi}{3} \right) \\ &= a^2 \end{aligned}$$

iii. $z^2 - az + a^2 = 0$

Let the roots be α and β .

$$\alpha + \beta = -\frac{b}{a} = a$$

$$\alpha\beta = \frac{c}{a} = a^2$$

$$z_1 z_2 = a^2 \quad (\text{part (ii)})$$

$$z_1 + z_2 = \bar{\omega} a + \omega a$$

$$= \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} + \cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right) a$$

$$= 2 \cos \frac{\pi}{3} \times a$$

$$= 2 \times \frac{1}{2} \times a$$

$$= a$$

$$\therefore z_1 \text{ and } z_2 \text{ are the roots of } z^2 - az + a^2 = 0.$$

$$\begin{aligned}
 \text{i. } 1+i &= \sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right) \\
 &= \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \\
 (1+i)^n &= (\sqrt{2})^n \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right)
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 1-i &= \sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right) \\
 &= \sqrt{2} \left(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right) \\
 (1-i)^n &= (\sqrt{2})^n \left(\cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4} \right)
 \end{aligned}$$

$$\begin{aligned}
 \therefore (1+i)^n + (1-i)^n &= (\sqrt{2})^n \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right) + (\sqrt{2})^n \left(\cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4} \right) \\
 &= (\sqrt{2})^n \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} + \cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4} \right) \\
 &= 2(\sqrt{2})^n \cos \frac{n\pi}{4} \quad \dots \text{as required}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. } (1+i)^n &= \binom{n}{0} + i\binom{n}{1} - \binom{n}{2} - i\binom{n}{3} + \binom{n}{4} + \dots + i^n \binom{n}{n} \\
 (1-i)^n &= \binom{n}{0} - i\binom{n}{1} - \binom{n}{2} + i\binom{n}{3} + \dots + (-1)^n i^n \binom{n}{n} \\
 (1+i)^n + (1-i)^n &= 2\binom{n}{0} - 2\binom{n}{2} + 2\binom{n}{4} + \dots + 2\binom{n}{n}
 \end{aligned}$$

Given n is divisible by 4, the last term = $+ \binom{n}{n}$

Using part (i):

$$2\binom{n}{0} - 2\binom{n}{2} + 2\binom{n}{4} + \dots + 2\binom{n}{n} = 2(\sqrt{2})^n \cos \frac{n\pi}{4}$$

$$\binom{n}{0} - \binom{n}{2} + \binom{n}{4} + \dots + \binom{n}{n} = (\sqrt{2})^n \cos \frac{n\pi}{4}$$

If n is a positive integer divisible by 4 then $\cos \frac{n\pi}{4} = \pm 1$,
depending on whether n is an even or odd multiple of 4.

$$\therefore \binom{n}{0} - \binom{n}{2} + \binom{n}{4} + \dots + \binom{n}{n} = (-1)^{\frac{n}{4}} (\sqrt{2})^n$$